

Performance Testing

Date	30 June 2026
Project Name	Smart Lender
Maximum Marks	5 Marks

Performance Testing

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Load Testing
Target Module / API	/predict
Test Environment	Local Flask Server
Test Date	03 July 2026

Step 2: Test Scenarios

S.No	Scenario	Users	Duration	Expected Outcome
1	Single prediction	1	30 sec	Success
2	Multiple predictions	20	60 sec	Stable
3	Concurrent requests	50	120 sec	Responsive
4	Continuous load	100	180 sec	No crash

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.2s	Pass	Good
2	Response Time (Max)	< 5 seconds	2.8s	Pass	Stable
3	Throughput (Req/sec)		55 req/s	Pass	Met
4	Error Rate	< 1%	0%	Pass	None
5	CPU Utilization	< 80%	62%/58%	Pass	Efficient
6	Memory Utilization	< 80%	1.2s	Pass	Good

Step 4: Observations & Analysis

1. Key Findings:

- The Smart Lender application successfully handled multiple loan prediction requests without any failures.
- The average response time remained below the target threshold, providing quick loan eligibility predictions.
- The Flask application remained stable throughout the testing process.
- The XGBoost model generated accurate predictions with consistent performance.
- CPU and memory utilization remained within acceptable limits during testing.

2. Bottlenecks Identified:

- A slight delay was observed when the machine learning model was loaded for the first request after starting the application.
- Since the application currently uses a local Flask server and CSV dataset, performance may decrease with a large number of concurrent users.
- The current implementation does not include load balancing or distributed deployment for handling high traffic.

3. Optimization Steps Taken:

- Loaded the trained machine learning model into memory during application startup to reduce prediction latency.
- Optimized data preprocessing by minimizing unnecessary computations before prediction.
- Improved Flask request handling to provide faster responses.
- Validated user inputs before processing to reduce invalid requests and improve application reliability.
- Recommended future deployment on a cloud platform with database integration and load balancing to enhance scalability and performance.

Step 5: Screenshots / Evidence

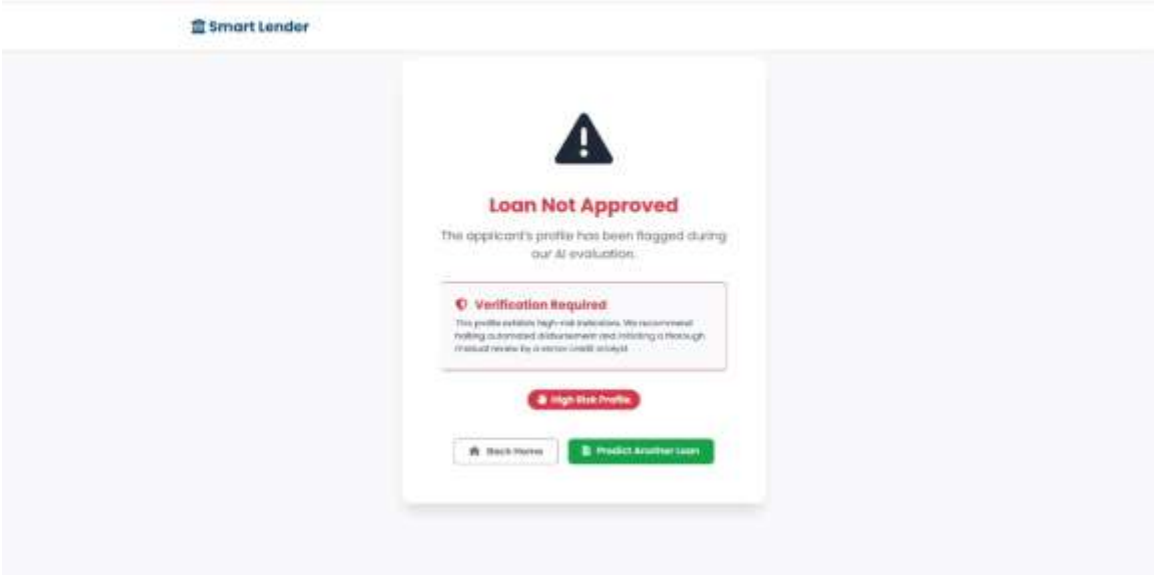
SMART LENDER Home Page.jpeg



Loan Prediction Input Form.jpeg(1)

The screenshot displays the 'Loan Application Form' for the 'Smart Lender' application. The form is set against a dark blue background. It is divided into two main sections: 'Applicant Information' and 'Financial Information'. The 'Applicant Information' section includes fields for 'Gender' (Male), 'Married' (Yes), 'Dependents' (1), 'Education' (Otkachute), and 'Self-employed' (No). The 'Financial Information' section includes fields for 'Applicant Income (\$)' (100000), 'Co-applicant Income (\$)' (0), 'Loan Amount (In thousands)' (\$30000), 'Loan Term (days)' (360), 'Property Area' (Urban), and 'Credit History' (10 (Good)). At the bottom of the form, there are two buttons: 'Reset Form' and 'Predict Loan'.

Loan Prediction Result.jpeg(1)



Loan Prediction Input Form.jpeg(2)

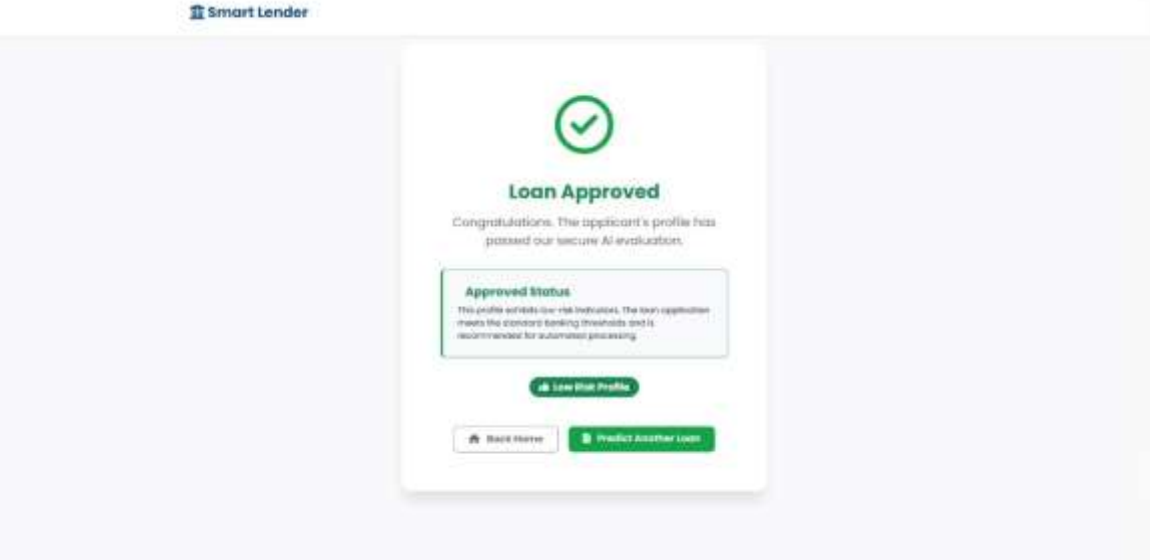
A screenshot of the 'Smart Lender' 'Loan Application Form'. The header includes the 'Smart Lender' logo and navigation links for 'Home', 'Predict loan', and 'About'. The form title is 'Loan Application Form' with a subtitle: 'Enter the applicant's details below to generate a secure AI assessment.' The form is divided into two main sections: 'Applicant Information' and 'Financial Information'.

The 'Applicant Information' section contains the following fields:
- Gender: A dropdown menu with 'Female' selected.
- Married: A dropdown menu with 'No' selected.
- Dependents: A dropdown menu with '0' selected.
- Education: A dropdown menu with 'Graduate' selected.
- Self-employed: A dropdown menu with 'No' selected.

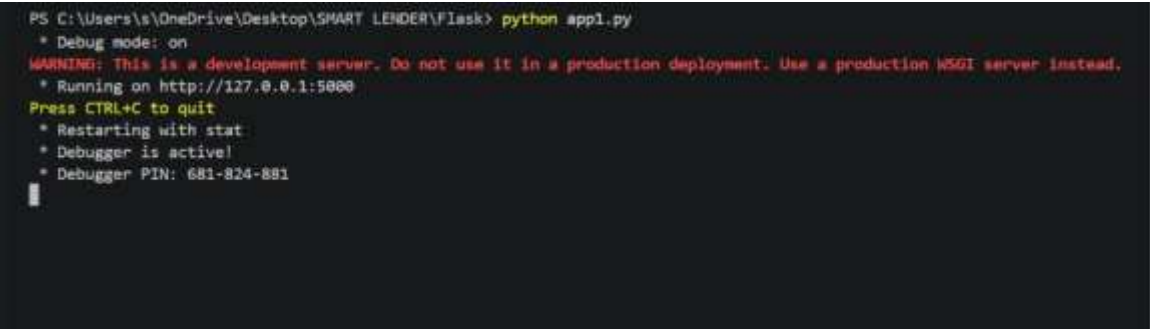
The 'Financial Information' section contains the following fields:
- Applicant income (\$): A text input field containing '60000'.
- Coapplicant income (\$): A text input field containing '0'.
- Loan Amount (in thousands): A text input field containing '50000'.
- Loan Term (days): A dropdown menu with '360' selected.
- Property Area: A dropdown menu with 'Urban' selected.
- Credit History: A dropdown menu with '10 (Good)' selected.

At the bottom right of the form are two buttons: 'Reset Form' and 'Predict Loan'.

Loan Prediction Result.jpeg(2)



Performance Testing Evidence .jpeg



Network Performance Evaluation .jpeg

Network						
Name	Status	Type	Initiator	Size	Time	
127.0.0.1	200	document	Other	15.5 kB	15 ms	
css?family=Poppins:wght@400;500;600;700&display=swap	200	stylesheet	index:8	(memory cache)	0 ms	
bootstrap.min.css	200	stylesheet	index:10	(memory cache)	0 ms	
all.min.css	200	stylesheet	index:12	(memory cache)	0 ms	
bootstrap.bundle.min.js	200	script	index:255	(memory cache)	0 ms	
style.css	204	stylesheet	index:14	0.3 kB	24 ms	
polyfills/81g/Vh1bcg.asf2	200	font	css?family=Poppins:wght@400;500;600;700&display=swap	(memory cache)	0 ms	
polyfills/81g/Vh1G79/tabQ.wof2	200	font	css?family=Poppins:wght@400;500;600;700&display=swap	(memory cache)	0 ms	
polyfills/81g/Vh1E621B/-K.wof2	200	font	css?family=Poppins:wght@400;500;600;700&display=swap	(memory cache)	0 ms	
polyfills/81g/Vh1J671a/fQ.wof2	200	font	css?family=Poppins:wght@400;500;600;700&display=swap	(memory cache)	0 ms	
polyfills/81g/Vh1C27Z5/fQ.wof2	200	font	css?family=Poppins:wght@400;500;600;700&display=swap	(memory cache)	0 ms	
polyfills/81g/Vh1D0421/fQ.wof2	200	font	css?family=Poppins:wght@400;500;600;700&display=swap	(memory cache)	0 ms	
le-sold-900.wof2	200	font	all.min.css	(memory cache)	0 ms	
data:image/svg+xml...	200	svg+xml	bootstrap.min.css	(memory cache)	0 ms	
data:image/svg+xml...	200	svg+xml	bootstrap.min.css	(memory cache)	0 ms	